The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2018-2019

MATHEMATICS FOR COMPUTER SCIENTISTS (COMP1017)

Time allowed ONE hour and THIRTY minutes only

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Questions on Sets and Functions.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) $|2^S| > |S|$.

(ii) $\emptyset \not\subset \emptyset$.

(iii) $\left|\{\{1,2,5\},\{3,4\},6\}\right| = 3$.

(iv) $\{0,2,12,36,80\} = \{y \in \mathbb{N} \mid \exists x \in \mathbb{N}((y = x^2 + x^3) \wedge (0 \le x \le 5))\}$.

(v) Ceiling is a function from integers to reals.

(b) Let $f, g\colon \mathbb{R} \to \mathbb{R}$. Given, $f(x) = x^3 + k$, $k \in \mathbb{Z}$ and $g(x) = x + l$, $l \in \mathbb{Z}$, work out the following.

(i) $f \circ g^{-1}$.

[6 marks]

(ii) What are the properties of $f \circ g^{-1}$? Explain your answer briefly.

[4 marks]

(c) Consider the following property of the floor function for real *x* and integer *m*

$$\lfloor x + m \rfloor = \lfloor x \rfloor + m$$

Prove this property (full marks are awarded for complete mathematical-type proof).

[10 marks]

2. Questions on Number Theory, Matrix and Relations.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) Consider a relation R where $\exists a, b, c: (a, b), (b, c) \in R \nRightarrow (a, c) \in R$. R is transitive.

(ii) The relation $>$ is symmetric.

(iii) The relation $\geq$ is reflexive.

(iv) Let g be greatest common divisor of integers m, n. Let $g = 2^{a_1} \cdot 3^{a_2} \cdot 7^{a_3} \cdots$ be written in its prime factorization form. If $a_i = 1$ and all others are 0, then g is a prime number.

(v) 173 is a prime number.

(b) An integer $p > 1$ is a prime number iff it is not the product of any two integers greater than one.

(i) Let $\mathbb{N}_2 = \{2,3,4, \dots\}$. Use $\mathbb{N}_2$ to mathematically formulate the statement above about prime number.

[6 marks]

(ii) Given $m = 252, n = 504$, calculate the two integers' prime factorizations. What is gcd(m, n)? Explain your answers.

[8 marks]

(c) Let $\mathbf{A}$ be a square matrix and $\mathbf{I}$ be the identity matrix. Given $\mathbf{A}$ such that $(\mathbf{I} - \mathbf{A})$ is nonsingular (invertible), prove that $\mathbf{A}(\mathbf{I} - \mathbf{A})^{-1} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{A}$. Hint: start with $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A}$.

[6 marks]

3. Questions on Counting and Probabilities.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) Given a set of integers $\{1, 3, 5, 7, 9, 11, 13, 15\}$, in the worst case one must select at least 6 integers from the set in order to guarantee that at least one pair of these integers add up to 16.

(ii) A bit string is of length 10. If the first two bits must be either 01 or 10, the total number of different bit strings satisfying such constraint is 2^{10}.

(iii) At a road junction the traffic light is changing between green and red every 30 seconds. A car reaching the junction would drive through if the traffic light is green and stop otherwise. We can say that the action of a car at the traffic light is a random variable.

(iv) The sample space of observing the outcome of tossing a coin twice is $\{HH, HT, TH, TT\}$, where H and T represent Head and Tail respectively.

(v) The probability of rolling a dice of 6 sides and observing a number that is less than 6 is 1.

(b) Ali designed a peculiar auto-typewriter that types 26 English characters independently and randomly. However, the letter 's' has the chances to be typed 5 times more than each of the other 25 letters which have equal chances to be typed out.

(i) In how many ways can the auto-typewriter type a word of length 5 such that no letter is repeated in the word?

[2 marks]

(ii) If there are exactly two consecutive 'x', how many different words of length 4 can the typewriter type? Show your work.

[3 marks]

(c) Given the auto-typewriter in (b), to make things even more peculiar, Ali also allowed letter 's', when typed, to be printed as 'a', 'w', 'd', or 'x', each of which with 10% chance, in addition to being printed as 's'. Other letters than 's' would always be printed the same as they are typed.

(i) Let X and Y be random variables that a letter is typed and a letter is printed, respectively. Let $P(X = x, Y = y)$ be the joint probability that a letter with value x is typed and a letter with value y is printed, where x and y are both in the set of 26 English characters. What is $P(X = '\text{a}', Y = '\text{s}')$ and $P(X = '\text{s}', Y = '\text{a}')$, respectively? Elaborate your answers.

[7 marks]

(ii) If you observed that the letter 'w' is printed, what is the probability that the letter 's' was typed? Elaborate your answer.

[8 marks]